



Review

Initiating Medications During Hospitalization and Strategies for Ensuring Linkage at Discharge for Patients With Opioid Use Disorder: A Scoping Review [☆]



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ABSTRACT

Hospitalization presents a critical opportunity to initiate medications for opioid use disorder (MOUD) and improve long-term outcomes for patients with opioid use disorder (OUD). While inpatient MOUD initiation significantly reduces mortality and relapse, many patients lack appropriate follow-up care after discharge. This scoping review synthesizes evidence from 52 studies on hospital discharge practices for patients with OUD initiated on MOUD to identify best practices that support continued treatment and recovery. Inpatient addiction consultation services, standardized protocols, and clinician education emerged as key facilitators of MOUD initiation. Transitional care strategies, such as bridge clinics, peer navigation, telemedicine, and structured discharge planning, were associated with increased outpatient linkage, reduced readmissions, and improved retention in treatment. Despite policy advances including X-waiver elimination, systemic barriers persist and disproportionately affect rural and minoritized populations. Multidisciplinary, patient-centered discharge pathways that integrate medical treatment with social support are critical. Effective linkage strategies must address both structural and individual barriers to care. We propose six pillars of MOUD continuity, including early initiation, warm handoffs, peer support, bridge care models, telemedicine integration, and attention to social determinants. Implementing these strategies is essential to closing care gaps and improving outcomes in the evolving landscape of MOUD treatment.

Introduction

The opioid epidemic remains a significant and preventable public health crisis in the United States, with opioid use disorder (OUD) contributing to widespread morbidity and mortality. Effective medications for opioid use disorder (MOUD), like methadone and buprenorphine, can reduce mortality by more than 50% when used long-term.¹ Hospitalization represents a critical opportunity to engage patients in evidence-based outpatient OUD treatment.²

Additionally, inpatient observation facilitates timely medication initiation for withdrawal management and enables streamlined transitions to outpatient OUD treatment. Initiating MOUD within 7 days of

an OUD-related hospital visit has been associated with a 37% reduction in the adjusted odds of fatal or nonfatal overdose at 6 months post-hospitalization.³ Despite this knowledge, there are significant challenges to initiating MOUD in a hospital setting, including insufficient training of prescribers, logistical hurdles (eg, electronic medical record [EMR] integration, development of new protocols), and limited access to addiction consultation services (ACSS).⁴

However, even patients who receive MOUD initiation during their hospital stay are at risk of inadequate follow-up care, leaving them vulnerable to return to opioid use and adverse outcomes post-discharge. The transition from hospital-based to outpatient care is a particularly challenging period for patients with OUD in early recovery. Fewer than

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half of hospitalized patients with OUD receive MOUD at discharge, and the rates of linkage to outpatient care are alarmingly low.⁵ Within the Veterans Health Administration, only 15% of veterans with OUD hospitalized for any diagnosis received MOUD during admission, and among those not previously on treatment, only 2% were newly initiated and linked to care post-discharge.⁶ Nearly 8% of patients with OUD die within 1 year of hospital discharge, most often from overdose, and many others cycle through emergency departments and rehospitalization due to preventable complications.² The absence of systematized transitional care not only jeopardizes long-term recovery but also contributes to avoidable healthcare utilization, morbidity, and mortality.

Recent evidence supports the utilization of structured discharge interventions to improve rates of MOUD continuation and reduce hospital readmissions.⁷ For instance, patients receiving ACS consultation were more likely to initiate MOUD during their hospital stay and maintain treatment post-discharge, with buprenorphine prescriptions increasing by 22% compared to historical controls.¹ In parallel with these clinical interventions, policy changes have also expanded access to treatment. In January 2023, the elimination of the X-waiver requirement through the Consolidated Appropriations Act enabled any DEA-licensed clinician to prescribe buprenorphine, removing a major barrier to MOUD access. This reform created new opportunities to initiate and continue treatment, particularly in rural and underserved areas where provider shortages have historically limited care.⁸

While policy changes have expanded access to buprenorphine, methadone prescribing remains largely restricted to federally regulated opioid treatment programs (OTPs), which require frequent in-person visits and reinforce geographic and systemic inequities.⁹ These limitations disproportionately impact rural and minoritized populations, who often face long travel distances to reach one of the unevenly distributed OTPs across the country.¹⁰ In response, recent regulatory updates such as the 3-day exception (the “72-hour rule”) now allow non-OTP providers to administer methadone temporarily, offering greater flexibility at the time of discharge. Although promising, these changes remain inconsistently implemented and highlight the ongoing policy challenges that must be addressed to ensure equitable care transitions for all patients with OUD.

With the increased access to MOUD in outpatient settings and evidence that inpatient initiation of MOUD improves outcomes, it is critical to establish effective pathways for linking inpatients to continued outpatient care to support long-term recovery upon discharge. Despite a growing body of research on MOUD, hospital protocols and discharge practices remain highly variable, with significant differences in implementation and outcomes. A recent scoping review focused broadly on care transition strategies for patients with substance use disorders (SUD), identifying interventions such as pre-discharge appointment scheduling, care navigation, and peer support, but noted a lack of standardization across systems and called for more rigorous evaluation.¹¹

This scoping review aims to synthesize current evidence to support best practices around linkage of hospitalized patients initiated on MOUD to outpatient care for continued treatment. It highlights the roles of bridge and transition clinics, community and peer support, patient incentivization and outreach in ensuring continuity of care. By integrating findings on discharge interventions and their impact on long-term outcomes, this review provides actionable recommendations to optimize MOUD care transitions in the evolving landscape of expanded treatment access.

Methods

This scoping review was conducted to identify and synthesize the breadth of existing literature addressing hospital discharge practices for patients with OUD requiring medications for treatment or MOUD. The review adhered to the Preferred Reporting Items for Systematic Reviews

and Meta-Analyses for Scoping Reviews (PRISMA-ScR) guidelines to ensure transparency and rigor throughout the search, selection, and synthesis processes.¹²

A scoping review was selected to address the 2-part research question: “What are the best practices and guidelines for initiating MOUD during hospitalization, and how do these practices contribute to successful care transitions and linkage to ongoing outpatient treatment?”. Given the heterogeneity in discharge interventions, patient populations, and outcomes reported across studies, a scoping review provided the flexibility to synthesize diverse findings and identify key themes, gaps, and opportunities for further research.

A comprehensive literature search was conducted using PubMed, Scopus, and Web of Science to identify peer-reviewed articles published in English within the last 10 years. The search strategy included the following terms and combinations: (“Opioid Use Disorder” OR “Opioid Addiction” OR “Opiate Addiction”) AND (“Medication Assisted Treatment” OR “Medication Assisted Therapy” OR “Opioid Agonist Therapy” OR “Methadone” OR “Buprenorphine” OR “Naltrexone”) AND (“Discharge” OR “Hospital Discharge” OR “Continuity of Care” OR “Long-Term Management” OR “Bridge Clinic” OR “Relapse Prevention”). Additional studies were identified through forward and backward citation tracking of key articles and by manually reviewing systematic reviews and meta-analyses for relevant references.

Studies were included if they focused specifically on patients with OUD discharged from hospital settings, examined interventions or practices related to hospital discharge, and/or reported post-discharge outcomes such as treatment retention, relapse rates, or continuity of MOUD. Quantitative, qualitative, and mixed-methods studies were eligible for inclusion. Articles were excluded if they addressed SUD broadly without specific attention to OUD, lacked relevance to hospital discharge practices, focused on countries other than the United States, were limited to interventions employed only during the COVID-19 pandemic, or did not report specific outcomes related to discharge interventions. Systematic reviews, meta-analyses, case reports, case series, editorials, commentaries, and non-peer-reviewed articles were also excluded.

Three independent reviewers (AD, KF, and one additional author) screened titles and abstracts for eligibility. Full-text articles were retrieved for all studies meeting inclusion criteria based on abstract review for further assessment. Discrepancies were resolved by majority consensus amongst the 3 reviewers. Data extraction was performed systematically via full text review to capture study characteristics such as authorship, publication year, country, and study design, as well as population details including sample size, demographics, and clinical characteristics. Data extraction also recorded intervention descriptions such as addiction consult services, transitional care strategies, and bridge clinics, along with reported outcomes including treatment retention, relapse rates, mortality, healthcare utilization, or any qualitative results descriptions. A single data extraction form was not utilized due to heterogeneity in study design, interventions, and outcome measures.

The intervention description was categorized into the following to evaluate impact on patient outcomes: (1) inpatient intervention, or (2) transitional care strategies. Outcomes were likewise synthesized narratively due to heterogeneity in study designs, interventions, and outcome measures. The analysis was structured to address 2 primary objectives: summarizing best practices for hospital-based MOUD initiation, discharge, and transitions-of-care, and assessing how these practices influence long-term outcomes, including treatment retention and relapse prevention.

Results

Literature Search Results

Our initial literature search yielded 373 articles, of which 85 met inclusion criteria after abstract review. Full text review resulted in the

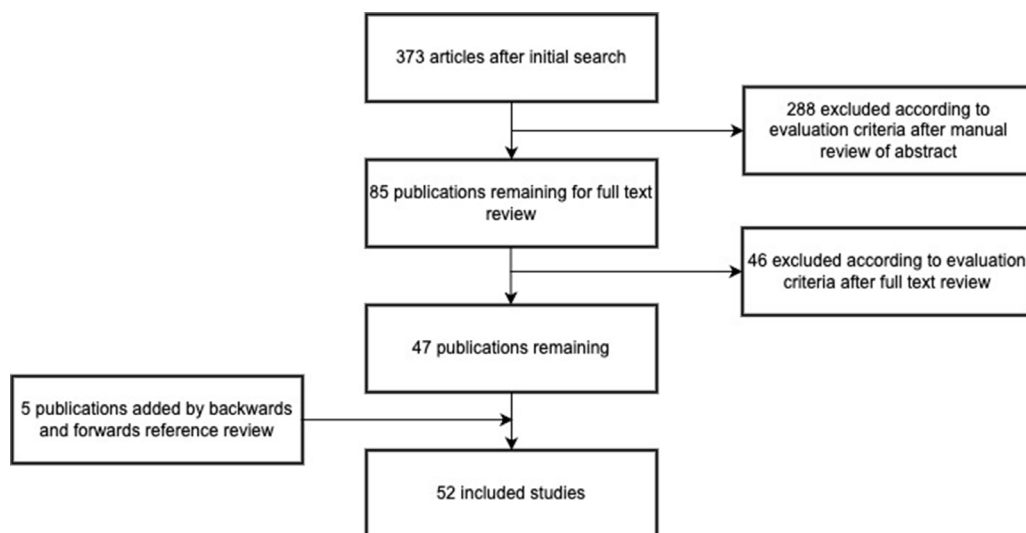


Figure 1. Article selection flow diagram. Flow diagram of the study selection process. The search strategy included the following terms and combinations: (“Opioid Use Disorder” OR “Opioid Addiction” OR “Opiate Addiction”) AND (“Medication Assisted Treatment” OR “Medication Assisted Therapy” OR “Opioid Agonist Therapy” OR “Methadone” OR “Buprenorphine” OR “Naltrexone”) AND (“Discharge” OR “Hospital Discharge” OR “Continuity of Care” OR “Long-Term Management” OR “Bridge Clinic” OR “Relapse Prevention”).

exclusion of 38 additional articles. Backwards and forwards citation review yielded an additional 5 studies that met inclusion criteria, for a total of 52 articles included in this scoping review (Figure 1).

The findings of this review are organized into 2 key categories: inpatient interventions and transitional care strategies. Most articles focused on strategies for initiating MOUD during hospitalization, emphasizing the importance of early intervention and addressing barriers such as clinician training and access to addiction consult services. Transitional care strategies, including bridge clinics, peer navigation, and telemedicine, were frequently highlighted as critical for ensuring continuity of care and reducing the risk of relapse or readmission following discharge. Together, these categories collectively emphasize the need for discharge practices that seamlessly link inpatient stabilization with community-based care to optimize recovery and reduce the risk of return to opioid use.

Inpatient Interventions

For many patients with OUD, particularly higher acuity patients who are not engaged in primary or outpatient care, their entry point into MOUD occurs during a visit to the hospital.¹³ Recent studies have focused on optimizing MOUD initiation during inpatient stays, and in our study a total of 32 out of 52 (61.5%) included studies specifically addressed inpatient interventions for patients with OUD to effectively bridge patients to long-term treatment. A variety of study designs met inclusion criteria: 13 (40.6%) were retrospective cohort studies, 11 (34.4%) were program evaluation studies, 5 (15.6%) were randomized controlled trials (RCTs), 1 (3.1%) was a survey-based study, 1 (3.1%) was a prospective cohort study, and 1 (3.1%) was clinical practice guidelines. Key findings of included studies on hospital-based interventions are shown in Table 1.

Inpatient care is a particularly salient entry point into treatment. Stein et al¹⁴ completed an RCT demonstrating that patients linked to OUD care through inpatient teams experienced significantly better outcomes than those who received standard withdrawal management. These outcomes included reduced opioid use, increased buprenorphine adherence, and higher outpatient follow-up rates. Similarly, Cushman et al¹⁵ found that initiating buprenorphine in hospitalized patients increased the likelihood of attending initial outpatient follow-up appointments.

Buprenorphine

Eight studies evaluated whether initiating buprenorphine during hospitalization improved MOUD prescription fill rates and outpatient follow-up. The included studies comprised 4 retrospective cohort studies, 3 program evaluation studies, and 1 RCT. Three of those studies reported increased rates of prescription fills within 60 to 90 days post-discharge.¹⁶⁻¹⁸ Noam et al¹⁹ found that induction of MOUD in the inpatient setting was associated with higher odds of post-discharge MOUD adherence (OR: 3.83; 95% CI: 3.06-4.81). Of the MOUD adherent patients, there were significant reductions in acute service utilization and opioid overdose over a 90-day post-discharge period. Across several included studies, rates of outpatient follow-up ranged from 12% to 50% over periods between 2 weeks and 2 years.¹⁹⁻²² Stein et al¹⁴ also reported increased rates of outpatient follow-up as well as reduced opioid use at 6 months post-inpatient treatment initiation.

Various system-level interventions have additionally been explored as a strategy to support the streamlined initiation of buprenorphine in the inpatient setting. Clifton et al²³ evaluated the implementation of a hospitalist-led program that developed new protocols to standardize OUD treatment and referrals to post-discharge care facilitated by a social worker, which led to a significant increase in MOUD use during hospitalization (from 36% to 57%) and buprenorphine prescription at discharge (from 2% to 20%). Kerins et al²⁴ described the efforts of a workforce to develop an OUD consult service, formulary revisions, creation of an embedded clinical pathway in the EMR, education for healthcare workers, outpatient partnerships, and pharmacy-led initiatives. These initiatives led to increased administration of MOUD over time, greater naloxone dispensing at discharge, and lower readmission rates among patients who received an OUD consult compared to those who did not. Finally, Thakrar et al²⁵ implemented a quality improvement program at a large academic center that focused on increasing resident education, which increased buprenorphine initiation rates from 10% to 24%.

Methadone

In addition to buprenorphine, methadone is another highly effective form of MOUD that may be utilized for patients with OUD during hospitalization.^{26,27} Shanahan et al²⁸ describe a single institution experience with methadone initiation and subsequent outpatient follow-up, with 82% of patients completing at least one visit at the outpatient clinic. In 2022, the FDA’s approval of a 3-day methadone exception for

Table 1
Key Findings of Included Studies on Inpatient Interventions (n = 32)

Study	Design	Findings
Stein et al ¹⁴	RCT	Evaluated the effectiveness of initiating buprenorphine treatment during short-term inpatient detoxification for OUD and providing linkage to office-based buprenorphine treatment post-discharge. Participation in linkage to office-based buprenorphine after discharge had lower illicit opioid use rates at several timepoints, and higher prescription buprenorphine use rates ($P < .001$) at all timepoint assessments.
Cushman et al ¹⁵	RCT	Linkage of injection opiate users to outpatient buprenorphine did not significantly decrease their injection opiate use frequency. Although linkage patients (70.6%) were significantly more likely ($P < .001$) to present to initial buprenorphine visits than detoxification patients (9.7%), there was no significance between the 2 groups at later timepoints of 1-, 3-, and 6-month follow-up. Suggested that additional strategies may be needed to sustain outpatient buprenorphine treatment.
Clifton et al ²³	Program evaluation	A hospitalist-led program was created and demonstrated a successful approach to develop protocols for standardized OUD treatment, implement evidence-based treatment for inpatients with OUD, and refer patients to post-discharge care facilitated by a social worker. Among all patients at the academic medical center with OUD, MOUD use during hospitalization and at discharge increased in the period after the program creation compared to the period before implementation ($P < .001$ for both).
Kerins et al ²⁴	Program evaluation	A multidisciplinary work group developed an OUD consult service, formulary revisions, education for health care workers, outpatient partnerships, and pharmacy-led initiatives and conducted 296 OUD consults, of which 103 consult patients (35%) received and were discharged with buprenorphine, and 118 patients (40%) were managed with methadone and linked to outpatient care.
Thakrar et al ²⁵	Program evaluation	Studied the impact of both clinical education and implementation of a team of residents and attendings waived to prescribe buprenorphine to initiate patients with OUD on buprenorphine maintenance. The rate of starting buprenorphine maintenance increased from 10% to 24% after the intervention, with a significant increase in rate (14.4%; 95% CI, 3.6%-25.3%; $P = .02$). The engagement in treatment at 30 days after discharge was unchanged (40%-46%). Survey responses of internal medicine residents demonstrated improved knowledge and comfort with buprenorphine.
Quaye et al ¹⁶	Retrospective cohort	There was a statistically significant increase in MOUD prescribing 3 months post-hospital discharge in patients who received MOUD at time of discharge (87.9% vs 40.0% $P = .002$). There was a significant increase in nonbuprenorphine opioid prescribing in patients who did not receive an MOUD prescription at time of discharge (24.2% vs 70.0% $P = .007$). There was a significant reduction in the incidence of non-MOUD opioid dispensing in patients prescribed MOUD at measured timepoints including time of discharge, 3, 6, and 12 months ($P = .007$, $P < .001$, $P < .001$ and $P = .008$). These findings support continuing buprenorphine as prescribed medication for OUD following hospital discharge.
Mosher et al ¹⁷	Retrospective cohort	Among 830 individual inpatient hospitalizations at VA hospitals, inpatient buprenorphine administrations in the final 36 hours prior to discharge were highly predictive of continued outpatient buprenorphine receipt (94.0%, compared to only 63.7% among those not receiving buprenorphine near the time of discharge).
Patel et al ¹⁸	Retrospective cohort	An anesthesiologist-led inpatient buprenorphine initiation effort effectively managed inpatient pain and facilitated linkage to outpatient buprenorphine providers in hospitalized patients with OUD. Five out of 7 patients started on buprenorphine for acute pain filled their prescriptions after discharge. Overall pain levels did not worsen during the first 24 hours after buprenorphine initiation, and all 7 patients remained on buprenorphine until hospital discharge without resumption of any other opioids.
Noam et al ¹⁹	Retrospective cohort	Following a pilot program to increase inpatient MOUD induction, of 2332 Medicaid adults diagnosed with OUD and discharged from 2 inpatient withdrawal management facilities, 493 started MOUD inpatient care (21.1%) with most initiating buprenorphine (65.5%). Induction of MOUD was associated with a lower likelihood of discharge against medical advice (OR, 0.49; 95% confidence interval [CI], 0.37-0.64), 30-day all-cause hospital readmission (OR, 0.61; 95% CI, 0.47-0.80), and higher odds of post-discharge MOUD adherence (OR, 3.83; 95% CI, 3.06-4.81). MOUD adherent patients had significant reductions in emergency department visits for short-term service utilization and opioid overdose in the 90 days after discharge.
Moriates et al ²⁰	Program evaluation	The implementation of a hospital-based opioid treatment team at a medical center showed increased post-discharge follow-up rates and retention up to 1 year following discharge. Strategies included initiation of buprenorphine therapy, coordination of outpatient care prior to hospital discharge, scheduling for follow-up at partner FQHC addiction treatment clinics, or scheduling with other outpatient or virtual providers. Additionally, all care-coordinated patients were provided with buprenorphine-naloxone prescriptions to bridge them to their first outpatient appointment.
Christian et al ²¹	Program evaluation	A hospitalist-led buprenorphine team was created to identify hospitalized patients with OUD, initiate buprenorphine in the inpatient setting, and provide bridge prescriptions and access to outpatient treatment programs. Of 132 patients who were administered buprenorphine therapy during hospitalization, 110 (83%) were bridged to an outpatient program, and 65 (59%) of those attended initial follow-up appointments with 24 (22%) still engaged at 6 months.
Fine et al ²²	Program evaluation	A volunteer physician-led intervention at a hospital evaluated 178 patients, with 47 patients initiated on buprenorphine while hospitalized. Sixty-seven patients were linked to primary care-based buprenorphine treatment, of which 29 (43%) attended an appointment at least 30 days after discharge. Retention rates of those patients seen at initial follow-up were 22 (76%) who returned at >60 days, 20 (69%) at 6 months, and 16 of 17 possible patients (94%) seen at 1 year.
Nosyk et al ²⁶	Retrospective cohort	The use of methadone for treatment of OUD was associated with a lower risk of treatment discontinuation compared with buprenorphine/naloxone.
	Practice guideline	Describes the ASAM National Practice Guideline for the Treatment of OUD to provide information on evidence-based treatment of OUD.
Shanahan et al ²⁸	Prospective cohort	A transitional opioid program for opioid-dependent drug users led to the identification of at-risk hospitalized out-of-treatment patients and provided interim opioid agonist treatment while offering linkage to a variety of treatment intensity options. 82% of enrolled patients presented to the program clinic post-hospital discharge, and 59% of enrolled patients received addiction treatment.
Bowman et al ²⁹	Program evaluation	Following the DEA exception to dispense methadone in acute care settings, a new program that created processes for methadone dispensing and educational materials was implemented and successfully demonstrated feasibility of expanding patient access to methadone at hospital and ED discharge.
Skogrand et al ³⁰	Program evaluation	Described the implementation of dispensing methadone at hospital discharge following the "72-hour rule" change, detailing implementation challenges such as interdisciplinary staff education and support from pharmacy champions, EHR capabilities, and establishment of supportive policies and practices.
Jakubowski et al ¹	Retrospective cohort	Participants that received addiction consult service (ACS) consultation were significantly more likely to have a discharge prescription for buprenorphine (OR, 0.17.22; 95% CI, 3.94-75.13) and new methadone administration at a methadone program within 30 days after discharge (7.0% vs. 0.0%, $P = .007$).
Trowbridge et al ³¹	Program evaluation	A new inpatient addiction consultation service was created to provide diagnosis, initiation of treatment medications, and discharge linkage consultations for patients with SUD in the inpatient setting. Of patients initiated on methadone, 76% were linked to methadone clinics; of patients initiated on buprenorphine, 49% were linked to a clinic; of patients initiated on naltrexone, 26% were linked to a clinic.

(continued on next page)

Table 1 (continued)

Study	Design	Findings
Brothers et al ³²	Program evaluation	Assessed the implementation of a trainee-organized hospital addiction medicine consultation service, demonstrating acceptance of the service by hospital providers and patients, feasibility of implementation, and a positive impact on patient care as determined by patient initiation on opioid agonist therapy (OAT) in the hospital and fulfillment of an outpatient prescription or attendance of first visit with an outpatient OAT prescriber.
Tran et al ³³	Program evaluation	A substance use intervention team comprised of interdisciplinary members was successfully implemented in a hospital to provide universal screening, inpatient consultation services, initiation of medications for SUD for appropriate patients, and medical and behavioral clinic visits to transition patients to long-term treatment.
Nordeck et al ³⁴	RCT	Most of the patients seen by a hospital SUD consultation service met criteria for severe OUD ≥ 6 DSM-5 symptoms and represented a vulnerable, high-risk patient population. Linkage to community-based OAT within 30-days post-discharge was associated with hospital-based buprenorphine initiation (Adjusted Odds Ratio [AOR], 3.86; 95% CI, 1.73-8.61; $P = .001$) and patient navigation intervention (AOR, 2.97; 95% CI, 1.60-5.52; $P = .001$).
Wakeman et al ³⁵	Retrospective cohort	A hospital-wide SUD initiative including an inpatient addiction consult team, low-threshold bridge clinic, recovery coaches, and office-based addiction treatment nurses led to 7036 unique patients being seen over the course of 5 years, with a variety of substance use disorders. Following the initiative, SUD pharmacotherapy was successfully initiated with 1623 patients receiving a buprenorphine prescription during the study period (42% of patients with OUD), 877 for oral naltrexone, and 163 for extended-release naltrexone. The mean length of continuous treatment episodes were longer with buprenorphine.
Ober et al ³⁶	RCT	Of the 38 patients that were randomized to meet with a newly implemented hospital inpatient addiction consult team, 97% met with the team and 89% received ≥ 8 of 10 intervention components. Patients assigned to the addiction consult team had higher odds of initiating medication during the inpatient stay (OR, 6.26; 95% CI, 2.38-16.48; $P < .001$) and being linked to follow-up care (OR, 5.76; 95% CI, 1.86-17.86; $P < .01$) compared to usual care patients ($N = 50$).
Ehrhard et al ³⁷	Retrospective cohort	The addition of a clinical pharmacist practitioner to an inpatient addiction triage team improved medication management and patient outcomes, and achieved both statistically and clinically significant improvements in MAUD/MOUD initiation rates prior to discharge compared with the historical control group (26.3% vs 4.0%, $P < .0001$).
Englander et al ³⁸	Retrospective cohort	Participants who received ACS consultation engaged in SUD treatment following hospital discharge more frequently than controls (38.9% vs. 23.3%, $P < .01$; AOR, 2.15; 95% CI 1.29-3.58).
Zavodnick et al ³⁹	Retrospective cohort	Following introduction of withdrawal guidelines for patients with opioid withdrawal at a hospital without an addiction consult service, use of OAT in patients increased significantly from 23.4% to 64.8% ($P < .001$). There was no associated change with patient-directed discharge, which remained at 14%, and 30-day readmissions increased from 12.4% to 15.7% ($P = .051$).
Jack et al ⁴⁰	Retrospective cohort	For patients who were started on buprenorphine during hospitalization, patients who had an encounter with a peer provider did not significantly differ in the rates of receiving a buprenorphine prescription (RR, 1.06; 95% CI, 0.74-1.51), hospital readmission (RR, 1.45; 95% CI, 0.80-2.64), or attendance at a buprenorphine follow-up visit (RR, 1.03; 95% CI, 0.68-1.57).
Suzuki et al ⁴¹	RCT	A peer recovery coach intervention for hospitalized adults with OUD receiving MOUD treatment showed no significant differences in the retention of participants in MOUD treatment at 6 months, readmission rates at 6 months, or the number of days until treatment discontinuation or hospital readmission.
Bhatraju et al ⁴²	Retrospective cohort	Among hospitalized patients with OUD who were seen by an addiction medicine consult service and initiated on buprenorphine, 30.5% were hospitalized for traumatic injuries, and 63.2% of those trauma patients were seen at follow-up within 30 days compared to 48.2% of nontrauma patients ($P = .16$).
Reif et al ⁴³	Retrospective cohort	When follow-up behavioral health services were received by patients with SUD within 14 days of discharge from an inpatient hospital stay or residential detoxification, the provision of MAT or residential treatment was associated with reduced risk of 90-day behavioral health admission.
Calcaterra et al ⁴	Survey	Characterized barriers to initiate in-hospital medications for OUD treatment. Clinicians reported that possible interventions such as treatment protocols to initiate medications for OUD, increased addiction specialist support and education, and post discharge treatment linkage could facilitate hospital-based OUD treatment provision.

bridging outpatient care enabled hospitals to explore inpatient initiation protocols. Two program evaluation studies described the integration of routine methadone initiation in an inpatient setting since this exception. Bowman et al²⁹ described a protocol implemented in 2 Baltimore hospitals, which included education and EMR integration, serving 36 patients within the first 3 months. Skogrand et al³⁰ similarly emphasized the importance of pharmacist involvement and integration, though they noted challenges with scaling methadone initiation as routine care.

Addiction Consult Service

ACSs have become an essential component of inpatient care for patients with OUD, particularly through initiation of medication and linkage of patients to outpatient care. Eleven of the included studies evaluated ACSs, consisting of 5 program evaluation studies, 4 retrospective cohort studies, and 2 RCTs. Overall, the results of 8 of these studies with patient cohort sizes ranging from 60 to 5000 consistently showed high rates of treatment initiation and linkage to outpatient clinics.^{1,23,24,31-35} For example, in an observational study of a newly formed ACS, Trowbridge et al³¹ found high rates of linkage for patients started on MOUD (49% for buprenorphine and 76% for methadone) after the new inpatient ACS began providing diagnostic, management, and discharge linkage consultations. In a secondary analysis of the NavSTAR trial, Nordeck et al³⁴ found that both hospital-based buprenorphine initiation by ACS (AOR, 3.86; 95% CI, 1.73-8.61) and the use of patient navigation ser-

vices (AOR, 2.97; 95% CI, 1.60-5.52) were independently associated with significantly increased odds of linkage to community-based opioid agonist treatment within 30 days post-discharge, demonstrating the synergistic value of medication initiation and transitional support strategies.

Ober et al³⁶ conducted a pilot RCT (Substance Use Treatment and Recovery Team, START), which demonstrated higher odds of MOUD initiation and outpatient follow-up among patients seen by ACS teams, finding that patients who were seen by an ACS were 5.76 times more likely ($P < .01$) to be linked to follow-up care. Ehrhard et al³⁷ reported further improvement in MOUD initiation rates by incorporating clinical pharmacist practitioners into ACS teams. Jakubowski et al¹ and Englander et al³⁸ similarly reported that patients engaged through ACSs were more likely to attend outpatient clinics and sustain MOUD adherence. These studies highlight the critical role of ACSs in facilitating a smooth care transition in order to ensure continuity of care and improve long-term outcomes.

Not all hospitals, however, have access to an ACS and must rely on preexisting structures to treat OUD in an inpatient setting. Zavodnick et al³⁹ developed a set of guidelines for medication initiation, patient directed discharge, and readmission for general hospitalists. The provider guidelines included specific information on dosing for both methadone and buprenorphine as well as discharge instructions; and the implementation of this intervention significantly increased MOUD initiation.

Other Included Inpatient Interventions

In addition to initiation of medical treatment, other studies focused on additional inpatient support that is associated with improved long-term outcomes. Jack et al⁴⁰ and Suzuki et al⁴¹ investigated the impact of peer recovery coaches on MOUD retention, but neither found statistical improvements in measures of treatment success including outpatient follow-up, readmission rates, or prescription pick-up.

Two studies focused on patient populations that face unique challenges requiring tailored inpatient interventions. Bhatraju et al⁴² reported that trauma patients were less likely to follow up after buprenorphine initiation, emphasizing the importance of effective care linkage. Although Medicaid patients are more susceptible to readmission, Reif et al⁴³ found that initiation of MOUD for patients with Medicaid reduced 90-day-readmission rates.

Despite these successes, barriers to inpatient MOUD implementation remain. Calcatera et al⁴ identified legal uncertainties, inconsistent protocols, and lack of discharge planning as major barriers in a survey conducted amongst hospitalists. These barriers can be addressed through ACS discharge practices such as warm handoffs and scheduled outpatient follow-ups, which provide structured support for MOUD initiation and discharge planning.

Transitional Care Strategies

Following inpatient intervention, it is essential to quickly transition patients to long-term care to improve treatment retention and decrease hospital readmissions. Transitional care pathways are designed to bridge the gap between inpatient and outpatient management, and include bridge clinics, telemedicine referrals, residential facilities, and other less common approaches. In our study, a total of 20 studies were included that focused on transitional care strategies. A variety of study designs investigating transitional care strategies met inclusion criteria: 7 (35.0%) were retrospective cohort studies, 4 (20.0%) were program evaluation descriptive studies, 3 (15.0%) were RCTs, 3 (15.0%) were qualitative descriptive studies, 1 (5.0%) was a prospective cohort study, 1 (5.0%) was a pre-post analysis of administrative claims data, and 1 (5.0%) was a survey-based study. Key findings of included studies on transitional care strategies are shown in Table 2.

Krawczyk et al⁴⁴ used a modified Delphi consensus process to develop recommendations for hospital-based transition strategies. Among the 45 experts who participated, the items ranked highest in both impact and feasibility included arranging for outpatient staff to meet the patient while in the hospital, scheduling outpatient appointments and telehealth visits prior to discharge, and coordinating integration of acute medical care with MOUD treatment. Other items, such as providing transportation to outpatient care, were deemed very impactful but less feasible for many hospital systems. Chutuaue et al⁴⁵ and Summers et al⁴⁶ explored the use of financial incentives to increase engagement in further treatment, with demonstrated improvement in patient outcomes, but this approach is not reasonable for many organizations. Another important consideration for hospitals is prompt scheduling of the first outpatient appointment. Roy et al⁴⁷ found that patients with zero to 1 day of wait time had 2.6 times higher odds of arriving at their appointment compared to those who waited 2 or more days. Tierney et al⁴⁸ identified stable housing and in-hospital methadone initiation as key predictors of linkage in a large retrospective study. In general, optimizing discharge processes is essential for prompt and successful linkage to outpatient treatment.⁴⁹

Bridge Clinics

Eight studies—comprising 1 randomized controlled trial (RCT), 5 observational noncontrolled studies, 1 pre-post analysis of administrative claims data, and 1 qualitative descriptive study—investigated the utility of bridge clinics in supporting care transitions. Bridge clinics, which are short-term, low-barrier medical services designed to provide immediate access to MOUD and supportive care for individuals transitioning

from inpatient or emergency settings, have also demonstrated significant success in transitioning patients to outpatient recovery services. However, it is important to note that in many of the cited studies, the patient populations were heterogeneous and not limited to individuals recently discharged from the hospital. Referral pathways commonly included emergency departments, outpatient providers, and self-referral, and outcomes were not stratified by referral source. This limits the ability to draw conclusions specifically about post-hospitalization transitions, although the model remains relevant for bridging gaps in care.

The recently published BRIDGE RCT from Marcovitz et al⁵⁰ found that patients referred to a bridge clinic were more likely to receive outpatient linkage, have more MOUD refills, and be less likely to overdose, although the bridge clinic referral did not reduce inpatient length of stay. Several studies from individual hospitals or small hospital networks also highlight the benefits of bridge clinics; however, these studies did not explicitly isolate outcomes for patients referred after hospital discharge. Taylor et al⁵¹ explored the use of the 72-hour methadone rule in a bridge clinic, which allows practitioners that are not a part of an OTP to prescribe methadone once a day for up to 72 hours while assisting the patient in finding a formal methadone treatment program. The researchers found that 87% of patients were linked to outpatient care and 58% were retained at 1 month, although patients were primarily emergency department or community-referred. Peckham et al⁵² reported that 65% of patients remained on extended-release buprenorphine 6 months after engaging with a Massachusetts General Hospital bridge clinic. Additionally, a recent pre-post study of a telemedicine bridge clinic found that 91% of patients filled a buprenorphine prescription within 30 days, and the intervention was associated with reduced unplanned care utilization and a 62% decrease in related per-member-per-month costs at 180 days.⁵³ In a qualitative study on bridge clinic experiences from Snow et al,⁵⁴ patients highlighted the clinic's accessibility, compassionate approach, use of peer support, and focus on harm reduction.

Bridge clinics have also demonstrated significant success in facilitating care transitions to office-based opioid treatment (OBOT) and sustaining long-term MOUD engagement. Casey et al⁵⁵ examined outcomes following treatment in a low-threshold bridge clinic and found that 78% of patients successfully transitioned to community-based care, with 84% still engaged in treatment and 68% taking MOUD at 22 months. Wake-man et al⁵⁶ reviewed follow-up and transfer to ongoing care outcomes amongst SUD patients seen at a low-threshold bridge clinic, of which 84% of visits were for OUD. In this study, however, only 38% of patients were retained at 60 days, with just 28% having documented transfer to care. Despite complications with long-term continuity of care, the authors noted that bridge clinics offering same-day treatment initiation and flexible scheduling achieved higher rates of initial engagement and retention.

In addition to general treatment success, some studies highlight other outcomes associated with bridge clinics. Kawasaki et al⁵⁷ described developing and scaling a bridge clinic in Pennsylvania to address rural needs, successfully recruiting 70 physicians for X-waiver training and integrating peer recovery specialists for improved patient outcomes. Aronowitz et al⁵⁸ interviewed leadership and staff of a Philadelphia-based bridge clinic, who emphasized the importance of leveraging existing infrastructure to reduce unnecessary burdens and employing expert Substance Use Navigators to address common patient concerns.

Peer Support

Peer recovery specialists, also referred to as peer navigators, are known to provide emotional support, practical guidance, and advocacy for individuals with OUD. These professionals have shown promising outcomes for linking patients to outpatient care in a variety of bridging contexts. Lennox et al⁵⁹ discussed the unique value of peer support workers, whose lived experiences provide credibility and build trust with patients. As part of a larger study on financial incentives, Summers

Table 2
Key Findings of Included Studies on Transitional Care Strategies (n = 20)

Study	Design	Key Findings
Krawczyk et al ⁴⁴	Survey	Established expert consensus on strategies to promote transition of care following opioid use disorder (OUD) hospitalization. Strategies related to coordination of outpatient care, including appointment scheduling, prior to hospital discharge ranked highest in impact and feasibility.
Chutuape et al ⁴⁵	RCT	Staff escort from inpatient detoxification to outpatient aftercare, along with financial incentives (worth US\$13.00) were associated with a higher percentage of participants (76%) completing outpatient aftercare intake, compared to solely financial incentive (44%) or solely a standard referral to outpatient aftercare (24%).
Summers et al ⁴⁶	RCT	Among persons living with HIV with co-occurring substance use disorder, receiving assistance with navigation to outpatient care along with financial incentives was associated with lower time to linkage of care.
Roy et al ⁴⁷	Retrospective cohort	Participants who had a wait time of 0-1 day between inpatient discharge date and outpatient appointment date were 2.6 times more likely to arrive at their appointment (95% CI 1.3-5.5) compared to patients with a wait time of 2 days or greater.
Tierney et al ⁴⁸	Retrospective cohort	Patients with stable housing and in-hospital methadone initiation were more likely to be retained in methadone maintenance therapy (MMT).
Marcovitz et al ⁵⁰	RCT	Participants referred to the bridge clinic in the BRIDGE trial were more likely to receive linkage to care for MOUD treatment (AOR, 2.37; 95% CI, 1.32-4.26), compared to those receiving usual care. There was no improvement in hospital length of stay between groups.
Taylor et al ⁵¹	Program evaluation	In a bridge clinic utilizing the 72-hour rule for methadone, 92% of methadone administration episodes had a plan for ongoing care in place within 72 hours. 87% of referrals to outpatient MOUD providers were linked to care and 58% were retained at one month.
Peckham et al ⁵²	Retrospective cohort	Six months after engaging with a bridge clinic, 65% of participants remained on extended-release buprenorphine (XR-BUP).
Lynch et al ⁶⁰	Pre-post analysis	Patients who engaged with a telemedicine bridge clinic initiated buprenorphine at high rates, remained in treatment over time, and experienced a reduction in unplanned care and healthcare costs.
Snow et al ⁵⁴	Qualitative descriptive	Interviewed participants that had engaged with a bridge clinic noted the accessibility of the clinic, compassion of the staff, and emphasis on harm reduction as positive aspects of the experience.
Casey et al ⁵⁵	Retrospective cohort	Of participants that transitioned from a bridge clinic into community-based substance use disorder (SUD) care, 78% reported successful connection to community-based SUD treatment, 84% remained in care of time of follow-up, and 68% reported taking medication for SUD at time of follow-up. The median duration of follow-up, from the last bridge clinic appointment to the study survey, was 22 months.
Wakeman et al ⁵⁶	Retrospective cohort	70% of bridge clinic visits resulted in engagement with outpatient care,. 38% were retained at 60 days, and 28% had documented care transfer.
Kawasaki et al ⁵⁷	Program evaluation	At 6-month follow-up of the Pennsylvania Coordinated Medication Assisted Treatment program, wait time was significantly shortened and 70 local physicians had received waiver training to prescribe buprenorphine.
Aronowitz et al ⁵⁸	Qualitative descriptive	Leadership and staff interviewed (n = 14) from a Philadelphia-based bridge clinic discussed the importance of utilizing existing infrastructure and forming relationships with outside stakeholders to better transition care.
Lennox et al ⁵⁹	Qualitative descriptive	Patients, healthcare providers, and peer support workers interviewed (n = 14) highlighted the importance of peer support workers in acting as a bridge to assist with overcoming system barriers and navigating transitions within the healthcare system.
Lynch et al ⁶⁰	Program evaluation	At least one prescription for buprenorphine was filled within 30 days for all OUD patients seen in a telemedicine bridge clinic, and 77% of patients filled 2 or more prescriptions.
Tofighi et al ⁶¹	Retrospective cohort	62.3% of patients seen in a telemedicine bridge clinic were provided same-day visits. 54.8% were referred to community treatment, and of those, 51.4% completed at least one community treatment visit.
Lynch et al ⁶²	Retrospective cohort	Patients who were referred for MOUD from telemedicine were 1.64 times more likely to attend their initial clinical appointment and 2.59 times more likely to be engaged with treatment at 30 days relative to patients referred from an emergency department.
Tassey et al ⁶³	Program evaluation	Researchers evaluated the OUD MEETS program, aimed at increasing buprenorphine and methadone initiation for hospitalized patients with OUD. 85% of patients successfully completed treatment at a skilled nursing facility (SNF) and 46% were linked to additional OUD treatment following the conclusion of SNF-treatment.
Klein and Seppala ⁶⁶	Prospective cohort	MOUD was effectively administered alongside standard 12-step based treatment. Patients that reported medication compliance at 1 and 6 months following residential treatment had significantly higher rates of abstinence than patients who did not report medication compliance.

et al⁴⁶ found that navigation to outpatient care significantly decreased time to linkage for patients with co-occurring HIV.

Telemedicine

Telemedicine linkage, which gained prominence during the COVID-19 pandemic, has become an accessible and flexible option for many patients. Telemedicine uniquely offers audio-only visits, which many patients prefer. Three observational noncontrolled studies investigated the use of telemedicine in facilitating transition of care. Lynch et al⁶⁰ reported that telemedicine bridge clinics facilitated buprenorphine initiation, with 96% of patients filling a prescription within 30 days and 77% filling 2 or more prescriptions. These clinics effectively engaged high-risk, vulnerable populations, including Medicaid-insured (62%) or uninsured (19%) individuals, and overcame barriers such as limited access to audiovisual technology by utilizing audio-only visits for 79% of patients. Tofighi et al⁶¹ similarly found that 50% of patients referred via telemedicine successfully transitioned to community-based treatment programs. In a 2024 study, Lynch et al⁶² compared ED to telemedicine-based referrals, finding that telemedicine referrals resulted in a 1.64 times higher likelihood of attending an initial outpatient appointment and a 2.59 times greater likelihood of retention at 30 days.

Other Transitional Care Strategies

For high-risk patients, completing the intake process to enroll in an OTP while in the inpatient setting may serve as a necessary intermediary before transitioning to outpatient or facility care. Tassey et al⁶³ piloted a program that partnered with 2 skilled nursing facilities (SNFs) and 2 outpatient OTPs. The program included embedding case managers, peer recovery specialists, and social workers early in the inpatient stay. This approach resulted in 85% of patients completing treatment at an SNF, with 46% linked to OUD treatment afterward. Once in a residential treatment program, multiple studies have demonstrated that MOUD is associated with increased treatment retention; however, there are conflicting outcomes on whether or not MOUD improves or reduces treatment completion.^{64,65} Finally, some patients opt to engage in 12-step based programs. Klein and Seppala found that 71% of patients in religious or 12-step-based programs were initiated on MOUD, and 73% attended at least one follow-up program after discharge.⁶⁶

Discussion

The long-term utilization of MOUD has proven to be a critical intervention in mitigating opioid-related morbidity and mortality. Inpatient

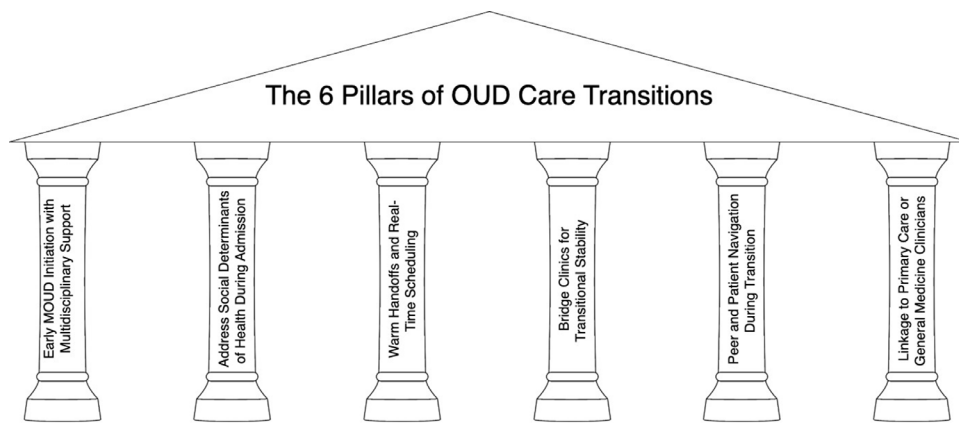


Figure 2. The 6 pillars of OUD care transitions. A visual depiction of 6 important pillars of successful OUD care transitions including Early MOUD Initiation with Multidisciplinary Support, Address Social Determinants of Health During Admission, Warm Handoffs and Real-Time Scheduling, Bridge Clinics for Transitional Stability, Peer and Patient Navigation During Transition, Linkage to Primary Care or General Medicine Clinicians.

settings represent a pivotal entry point for initiating MOUD. However, significant challenges remain in ensuring continuity of care from inpatient settings to outpatient management.

This review aims to synthesize current evidence on best practices for the treatment of patients initiated on MOUD during hospitalization, including ACS consultation and transition to outpatient treatment. Bridge and transition clinics, community and peer support, and patient outreach and incentivization are important supports for patients to mitigate drop-off between various stages of MOUD treatment. This article aims to address critical gaps in the literature and provide actionable recommendations for processes designed to optimize MOUD care transitions in the current landscape of expanding treatment access and unmet need.

The timely initiation of MOUD was a consistent theme across literature. Additional findings emphasize the need for clinician training and the development of local protocols to enhance readiness for MOUD initiation.⁶⁷ The integration of ACSs has also been shown to address logistical and clinical barriers during inpatient care, increasing the likelihood of successful MOUD initiation and follow-up care. Additionally, ACSs contribute to clinician education and capacity building within hospital settings, addressing discrepancies in addiction education across medical specialties.⁶⁸ By training inpatient teams and providing decision support, ACSs help to standardize care protocols and reduce variability in treatment practices.⁶⁹ This not only improves immediate patient outcomes but also fosters a culture of evidence-based care for OUD within hospital systems.

Despite these advancements, inpatient interventions alone are insufficient without a deliberate focus on care transitions. Efforts must prioritize not only medication initiation but also the development of sustainable pathways for outpatient engagement. The review highlights several evidence-based strategies for optimizing care transitions. A key function of ACSs is coordinating discharge plans that include warm handoffs to outpatient providers. By maintaining relationships with community-based programs, bridge clinics, and primary care providers, ACSs can provide cohesive transitions to long-term care. Peer navigation specialists have emerged as an impactful approach, demonstrating improved linkage rates and reduced time to outpatient care when peer recovery coaches were integrated into the care process. The role of lived experience in reducing stigma and improving patient trust further underscores the value of peer-based models.

Telemedicine has gained significant traction as a solution to geographic and logistical barriers in transitional care, particularly during the COVID-19 pandemic. Telemedicine models, including audio-only visits, have maintained accessibility for patients in rural or underserved areas. Bridge clinics, designed to provide short-term stabilization, have also demonstrated success in increasing outpatient linkage and reducing overdose risk. However, challenges such as limited scalability and funding constraints persist, requiring innovative solutions to broaden their reach.

Despite regulatory efforts such as the elimination of the X-waiver and the introduction of the 72-hour methadone rule, uptake of these

policies has been uneven and retention in OUD treatment remains challenging across the care continuum. Institutional barriers including limited clinician training, lack of standardized protocols, and insufficient infrastructure have impeded widespread implementation. MOUD initiation and retention are further influenced by a range of system-, clinic-, and patient-level factors, including geographic access, socioeconomic disparities, and co-occurring medical or psychiatric conditions. Patients initiated on buprenorphine during hospitalization consistently demonstrate higher adherence to treatment plans, although retention often declines over time.⁷⁰ Geographic barriers are particularly pronounced for methadone treatment, which requires daily attendance at federally regulated OTPs.⁷¹ Studies have shown that longer travel times to methadone clinics are inversely related to retention, highlighting the need for policies that expand clinic availability and address systemic inequities.⁷² In contrast, buprenorphine offers greater flexibility through integration into primary care and telemedicine models, enabling broader access and improved convenience for patients. Nevertheless, a recent quasi-experimental study of Medicaid beneficiaries in rural communities still demonstrated that MOUD initiation with either buprenorphine or methadone was associated with sustained reductions in emergency department visits and inpatient hospitalizations over a 30-month period.⁷³ Addressing these barriers requires coordinated efforts across healthcare and social service systems, including improved training for clinicians, expanded telemedicine infrastructure, and financial support for patients with limited resources.

In summary, this review highlights the essential components of effective MOUD treatment initiation and continuation, described in detail in Figure 2. This figure highlights the "6 Pillars of MOUD Treatment", which outline essential strategies for optimizing care including early MOUD initiation, addressing social determinants of health, ensuring warm handoffs, leveraging bridge clinics, peer navigation, and telemedicine to enhance continuity and accessibility. These pillars emphasize a multidisciplinary, patient-centered approach to support successful transitions from inpatient care to long-term recovery. Specific pathways that incorporate each element of best practices are illustrated in Figure 3, with distinct pathways outlined for patients initiated on buprenorphine or methadone. This figure outlines the care pathways for MOUD treatment, emphasizing a seamless transition from inpatient initiation, where social determinants of health and OUD diagnosis are addressed, to transitional care. Key components include tailored treatment initiation, peer navigation, bridge clinics, primary care follow-up, and telemedicine integration. Altogether, these findings emphasize the importance of integrating evidence-based strategies across the continuum of care to support patients with OUD.

Although this scoping review used rigorous search methodology that followed PRISMA-ScR guidelines, it does have limitations. First, it is possible that some relevant studies were not identified using the predetermined search terms. The use of forward and backward citation searching as well as including citations from review articles helped limit this problem. Additionally, there is the risk of publication bias in that it is more

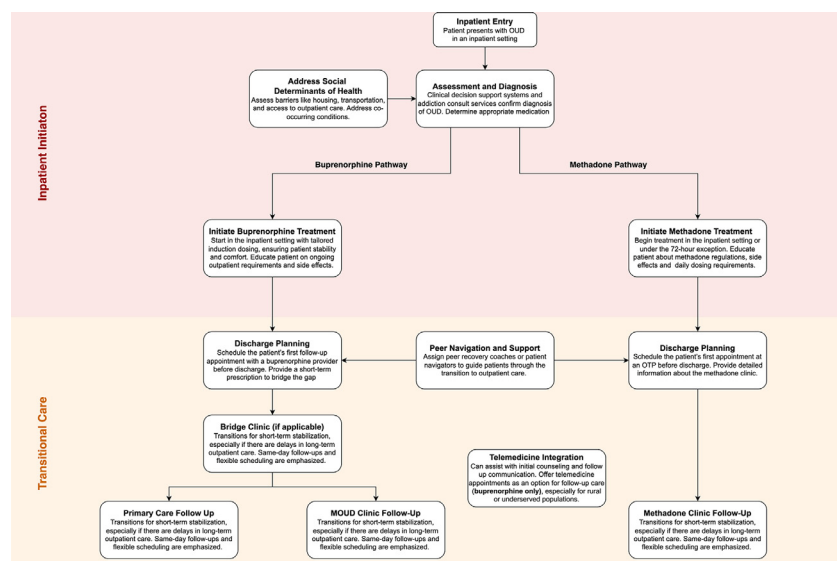


Figure 3. Ideal pathway for MOUD treatment. Flow diagram outlining the care pathways for MOUD treatment, from inpatient initiation to long-term outpatient management. The process begins with inpatient entry, where social determinants of health are assessed, and a clinical diagnosis of opioid use disorder (OUD) is confirmed. Patients are then guided through either the buprenorphine or methadone pathway, with tailored initiation of treatment, education, and discharge planning. Transitional care includes peer navigation, telemedicine integration, and the use of bridge clinics when applicable.

likely that positive studies were published. Furthermore, the heterogeneous endpoints of the included studies made quantitative synthesis of studies challenging.

Variability in MOUD accessibility and discharge practices across healthcare settings limits the generalizability of current findings, particularly in resource-constrained environments. While policy reforms such as the elimination of the X-waiver and the 72-hour methadone rule offer expanded access, their implementation remains uneven and is still being evaluated. Future research should investigate why some providers and institutions have not adopted these changes, including barriers related to training, institutional policy, or provider attitudes toward MOUD.

Given that many hospitals lack addiction consult services, future efforts should also focus on developing standardized discharge protocols that generalist providers can implement across diverse settings. Expanding peer recovery programs also holds promise, as navigators with lived experience, in limited data, have been shown to enhance patient trust, improve linkage rates, and reduce stigma. Similarly, initiating 12-step engagement during hospitalization may promote abstinence and provide ongoing social support that complements MOUD-based treatment. Ongoing research is needed to evaluate how these scalable, low-barrier models influence long-term treatment retention, particularly when integrated into transitional care.

Conclusion

This review underscores the need for a comprehensive, patient-centered approach to MOUD treatment that spans inpatient initiation and transitional care. Discharge planning that emphasizes linkage to outpatient care is critical to ensuring continuity of treatment and reducing the risk of return to opioid use or other adverse outcomes. By addressing structural inequities, leveraging innovative care models, and fostering multidisciplinary collaboration, healthcare systems can significantly improve access, retention, and outcomes for individuals with OUD. Future research should prioritize evaluating the scalability of transitional care strategies, the impact of policy changes like the X-waiver elimination, and the influence of these novel care delivery models on patient outcomes and long-term treatment retention.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Katherine Welter, MD served as a key advisor for Indivior.

CRedit authorship contribution statement

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